Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

AI Software Engineer with 6+ years building production AI systems, ML pipelines, and distributed infrastructure. At Intuit, designing and deploying AI infrastructure using the Anthropic Messages API and Agent SDK -- tool orchestration, structured output, input validation, guardrails, and responsible AI evaluation frameworks integrated into production financial systems at scale. Built AgentProbe, a multi-provider AI inference infrastructure with cost-aware routing and automated evaluation pipelines. Strong Python, distributed systems, cloud infrastructure (AWS), and agent orchestration (LangGraph). Ongoing investment in AI skill development: prompt engineering, context optimization, agentic workflow design, and emerging AI system architectures. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Python / AI Infrastructure | Agent Orchestration (LangGraph) | Messages API / Agent SDK | Prompt & Context Engineering | ML Pipelines (scikit-learn, XGBoost) | Responsible AI / Guardrails / Evals | Distributed Systems / Microservices | AWS (Lambda, S3, RDS, SQS) | Kubernetes / Terraform (IaC) | Docker / CI/CD (GitHub Actions) | Observability (Prometheus, OpenTelemetry) | PostgreSQL / Advanced SQL | Async Processing (RabbitMQ) | REST APIs / FastAPI

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer -- AI Systems & Infrastructure -- Virginia, USA

- Build **AI system infrastructure** using the Anthropic Messages API and Agent SDK: tool use, context engineering, structured output, authn/authz for model plugins, and multi-step agent orchestration integrated into production financial data workflows.
- Design and implement **responsible AI evaluation frameworks**: golden-dataset regression harnesses, bias-aware output validation, quality and accuracy reviews, and guardrails to enforce safe AI behavior before every release.
- Build **ML pipelines** (scikit-learn, XGBoost) from feature engineering through prediction scoring APIs; instrument with Prometheus and OpenTelemetry for production AI observability.
- Integrate AI tools to optimize development workflows; demonstrated efficiency gains through AI-assisted development (Claude Code) with rigorous review and testing before shipping.

Entrata -- Residential & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer -- Multi-tenant platform, 100K+ monthly active users

- Built **distributed data processing infrastructure** (RabbitMQ async, idempotency guarantees, dead-letter handling) supporting real-time eligibility and rent update pipelines across thousands of tenant records at scale.
- Designed and maintained **ML-adjacent decisioning infrastructure**: rule-based eligibility engine with AMI threshold calculations, income verification, and confidence-based routing -- the same pattern as production recommendation and ranking systems.
- Owned end-to-end data pipelines processing 1.6M+ records; optimized 200+ SQL queries (30% performance gain); led AWS S3 migration and PHI encryption covering 100% of resident data.
- Built Python automation reducing manual workload 80%; reviewed 300+ PRs; mentored junior engineers on distributed systems, data pipeline architecture, and engineering quality standards.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer -- Pune, India

- Built backend modules (Python, Java, PostgreSQL) with data pipelines processing 1.6M HUD/Tax Credit records; implemented role-based access control and multi-tenant data isolation.

PROJECTS

AgentProbe -- Multi-Provider AI Inference Infrastructure

Python · FastAPI · Provider Routing · Cost Management · CI/CD · Evals

15-endpoint FastAPI service routing across five LLM/ML inference backends with cost-aware model selection. Scoring engine evaluates output quality on five metrics; automated CI regression against golden datasets. Directly addresses AI infrastructure problems: observability, cost management, flexible model adoption, and eval-driven quality assurance.

github.com/venkatalakshmiparimala/AgentProbe

Leasel -- 8-Agent AI Orchestration System

Python · LangGraph · Agent SDK · FastAPI · Responsible AI

8-agent LangGraph orchestration system with triage, retrieval, rule evaluation, confidence scoring, and human escalation. Responsible AI design throughout: defined automation boundaries, bias-aware evaluation, and LLM-as-judge CI (93.6% precision, 87.9% F1 across 1,240 test cases).

github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR -- Clinical AI Infrastructure

Python · Flask · REST APIs · RBAC · Docker

RESTful AI system integrating LLM-based clinical NLP with FHIR data infrastructure. Fine-grained access control, containerized deployment, and automated testing -- regulated-domain AI infrastructure pattern.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017